

43

Myopathic Disorders

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SUMMARY PEARLS

- Myopathies are a diverse group of diseases with symptoms of weakness, cramps, and exercise intolerance. The two main categories are hereditary myopathies and acquired myopathies.
- Symmetric proximal weakness without sensory loss is the most common presentation. Specific myopathies such as facioscapulohumeral muscular dystrophy or inclusion body myositis have asymmetric involvement of select muscle groups.
- Diagnostic testing for myopathic diseases includes electromyography and nerve conduction studies, ultrasonography and/or magnetic resonance imaging, muscle biopsy, laboratory testing, exercise testing, and genetic testing.
- Causes of acquired muscle diseases are highly variable and may include autoimmune, endocrine, toxic, and infectious etiologies.
- Neuromuscular weakness leading to restrictive lung disease is one of the most serious manifestations in myopathic disorders, and management typically requires collaboration with a respiratory therapist and a pulmonologist.
- Strength and mobility assessments should be conducted at each visit for patients to evaluate the need for mobility aids and adaptive equipment.
- Pain is common in individuals with muscle diseases due a variety of factors, including musculoskeletal dysfunction, altered biomechanics, and joint contractures, requiring comprehensive evaluation and treatment strategies, including medications, therapies, imaging, and procedural interventions.

Introduction

Myopathies are a diverse group of diseases characterized by weakness caused by dysfunction of muscle fibers. The most common symptoms are weakness, cramps, rhabdomyolysis, and exercise intolerance. The pathophysiology of myopathies involves disruption of the muscle tissue integrity or internal metabolic processes, which can occur by either a hereditary or an acquired process. Hereditary myopathies can present in childhood, adolescence, or adulthood. Genetic mutations lead to deficiencies of the muscle architecture (muscular dystrophies), energy storage/processing (metabolic myopathies), or electrolyte channels (channelopathies). Acquired myopathies can be caused by a variety of causes, including autoimmune, toxic, endocrine, infectious, or medication related. Although the differential is broad, accurate diagnosis is important because many of these disorders have disease-modifying treatments.

Most myopathic disorders show a symmetric, proximal pattern of weakness, although there are key exceptions where weakness is asymmetric and preferentially affects certain muscle groups. Certain myopathic disorders cause dysfunction of other organ systems. Weakness of the diaphragm and respiratory muscles can cause restrictive lung disease, and weakness of the oropharyngeal muscles result in dysarthria and dysphagia. Cardiomyopathy and conduction abnormalities are found in many hereditary muscular

dystrophies. The ideal treatment of these patients combines medical therapies, multidisciplinary care, and rehabilitation management. The physiatrist is uniquely equipped to provide comprehensive care for this population.

History

A comprehensive history is essential for timely diagnosis and selecting the appropriate diagnostic tests. Myopathies typically involve motor impairment without sensory symptoms.

Determining the age of onset and rate of progression are crucial when differentiating between genetic and acquired myopathies. Most genetic myopathic disorders are present in childhood, adolescence, or by the third decade of life, whereas acquired myopathies can present at any age. Acquired myopathies usually present more rapidly with progressive weakness over a period of weeks or months, whereas hereditary myopathies can have a more indolent course over a period of years or decades. Episodic bouts of weakness with normal strength in between points toward a metabolic myopathy or periodic paralysis.²³ A birth and developmental history, including motor and cognitive developmental milestones, should be assessed. Hypotonia at birth may indicate a myopathic process, but spinal muscular atrophy and hereditary neuropathies are also considerations. For adult patients, the provider should also ask whether the patient had any issues with

physical activity and keeping up with peers in their youth, particularly in strength and endurance events. Muscle cramping and pain are often present. A history of rhabdomyolysis may also be present, and the provider should ask if there have been any episodes of dark red (cola color) urine.

Recognizing the pattern of weakness is essential. The majority of myopathic disorders, both acquired and hereditary, showed a proximal pattern of weakness preferentially affecting the shoulder muscles and pelvic girdle. This pattern holds true for the most common genetic myopathies, such as Duchenne muscular dystrophy (DMD) and limb girdle muscular dystrophy (LGMD). Most acquired myopathies, such as dermatomyositis, statin-induced myopathy, and steroid myopathy, also have this proximal pattern of weakness. Facioscapulohumeral muscular dystrophy (FSHD) presents with asymmetric weakness in the face, scapula, elbow flexors, elbow extensors, hip flexors, and dorsiflexors. Inclusion body myositis (IBM) is the most common myositis in adults over age 65 years and characteristically affects the long finger flexors and knee extensors.

A broad review of systems needs to be conducted for extramuscular symptoms. Respiratory muscle weakness can present with dyspnea on exertion and difficulty breathing when lying flat (orthopnea). Late-onset Pompe disease has significant diaphragm weakness, and this may be the presenting symptom. A cardiac evaluation for chest pains, palpitations, and swelling should be conducted. Cardiomyopathy and cardiac conduction abnormalities are frequently found in patients with myotonic muscular dystrophy, DMD, Emery-Dreifuss muscular dystrophy (EDMD), and select LGMD. The patient should be assessed for any difficulties with speech, swallowing, or saliva management. Dysphagia can be seen in IBM and oculopharyngeal muscular dystrophy (OPMD).²³

Exercise intolerance and fatigue are associated with metabolic myopathies, although these may be present in other myopathies as well. The type of exercise intolerance can be critical in determining the diagnosis. For example, glycogen storage disorders, such as McArdle disease, are characterized by muscle stiffness and weakness immediately after high-intensity exercise. Conversely, fatty acid oxidation defects, such as carnitine palmitoyltransferase II (CPT-II) deficiency, show exercise intolerance to low-intensity, long-endurance exercises.

The past medical history and medication history should be carefully reviewed. A history of autoimmune diseases, cancer, or skin manifestations can point toward inflammatory myopathy. Hyperthyroidism or calcium deficiency can also lead to myopathy. The most common medication-induced muscle disorders include statins, steroids, and colchicine. A detailed listing of inherited and acquired myopathic conditions is located in [Table 43.1](#).

Family History

Many myopathic disorders are hereditary in nature, and a detailed family history is necessary. Not all patients or parents can identify if a family member has a history of “muscular dystrophy.” Asking if patients have any family members who have an unusual gait, use braces or a wheelchair, or have a history of cardiac issues from a young age can be helpful. Completing a family pedigree may also provide insight into transmission of disease. Duchenne and Becker muscular dystrophies are X-linked recessive disorders that affect only males from the maternal side. Autosomal dominant disorders, such as FSHD, have a 50% chance of affecting the offspring,

regardless of sex, and many family members can be affected. In autosomal recessive disorders, such as LGMD, both parents are usually unaffected carriers, and the offspring have a 25% chance of getting the disease. In these cases, there may be no other family members involved or it may be a sibling that is affected.

Physical Exam

A comprehensive examination is essential to diagnosis. Manual muscle testing should be completed bilaterally in key muscle groups proximally and distally. Proximal weakness preferentially affecting shoulder abduction, external rotation, and internal rotation as well as hip flexion, hip abduction, and hip adduction is the most common pattern of weakness in myopathies. This is the pattern seen in DMD, Becker muscular dystrophy (BMD), and LGMD, as well as inflammatory myositis, statin-induced myopathy, steroid myopathy, and critical illness myopathy (CIM). Selective weakness of the finger flexors and knee extensors is seen in IBM. When testing finger flexors, have the patient place the hand supine and stabilize the distal interphalangeal joint to isolate the long finger flexors. Distal predominant weakness is rarer and can be seen in dysferlinopathy (LGMD type 2B), myofibrillar myopathies, and other late adult-onset muscular dystrophies.

Inspection of muscle bulk and appearance can be helpful. Atrophy of select muscle groups can point toward a certain diagnosis. Scapular winging is seen in FSHD, LGMD-2A, and LGMD-2C,D ([Fig. 43.1](#)). Pseudohypertrophy of the calves is seen in DMD ([Fig. 43.2](#)). EDMD and Ullrich congenital myopathy characteristically have elbow flexion contractures. Many of the congenital myopathies and congenital muscular dystrophies result in abnormal development in utero, leading to arthrogryposis (contractures) at birth.

Facial weakness can be seen in myotonic muscular dystrophy and FSHD. Weakness in ocular muscles or ptosis can be seen in FSMD and OPMD ([Fig. 43.3](#)). Macroglossia (enlargement of the tongue) can be seen in Pompe disease and amyloidosis. A high-arched palate or dental malocclusion can be seen in congenital myopathies and congenital muscular dystrophies. Patients with myotonic muscular dystrophy type I often have a characteristic hatchet-face appearance, tenting of the upper lip, temporal wasting, and frontal balding ([Fig. 43.4](#)).

Sensory examination should be intact in most myopathies. Reflexes are either normal or diminished based on the amount of weakness. Myotonic muscular dystrophies and channelopathies may present with grip myotonia. This can be assessed by having the patient maximally close both fists for 10 seconds and then relax them. Grip myotonia is observed if there is delayed relaxation. Percussion myotonia involves tapping the thenar eminence with a reflex hammer and assessing for involuntary contraction ([Fig. 43.5](#)).

If appropriate, the patient's gait and functional movements should be also evaluated. A “myopathic gait” is characterized by waddling of the hips from side to side, lumbar lordosis, and sometimes toe walking. Patients compensate for hip extension weakness by arching the lumbar spine posteriorly (lordosis) to keep the center of gravity over the pelvis. Similarly, they compensate for hip abduction weakness by leaning the trunk of the body over the standing leg during swing phase, which is also known as a compensated Trendelenburg gait ([Video 43.1](#)). Core and proximal hip strength can be further tested by asking the patient to get up out of a chair with the arms crossed or to perform squats.

TABLE 43.1**Inherited and Acquired Myopathies and Their Common Weakness Patterns and Associated Conditions**

	PREDOMINANT WEAKNESS PATTERN																				
	Proximal > Distal	Dist. Symmetric	Dist. Asymmetric	Scapuloperoneal	Dist. Lower Limb	EOM	Periocular Involve	Bulbar Dysfunction	Contractures	CNS Involvement	Acute Weakness	Hearing Involvement	Exercise Induced	Cardiac Involvement	Resp. Involvement	Myotonia/Cramps	Muscle Pain	Myoglobinuria	CPK > 1000	Immune Mediated	Inflammatory
Inherited Myopathies																					
Muscular Dystrophies																					
Duchenne muscular dystrophy	X								X	X				X	X			X*	X		
Becker muscular dystrophy	X								X	X				X	X	X		X	X		
Limb-girdle muscular dystrophy	X									X					X				X		
Congenital muscular dystrophy	X								X	X				X	X						
Facioscapulohumeral muscular dystrophy			X	X			X					X			X						X
Oculopharyngeal muscular dystrophy	X					X	X	X													
Emery-Dreifuss muscular dystrophy			X	X					X					X	X						
Dystrophic Myotonias																					
Myotonic muscular dystrophy type 1	X						X			X				X	X	X					
Myotonic muscular dystrophy type 2	X						X			X				X	X	X					
Nondystrophic Myotonias (Channelopathies)																					
Myotonia congenita		X											X			X					
Hyperkalemic periodic paralysis	X										X		X	X		X					
Paramyotonia congenita	X	X									X		X			X					
Hypokalemic periodic paralysis	X										X		X	X		X					

Continued

TABLE
43.1

Inherited and Acquired Myopathies and Their Common Weakness Patterns and Associated Conditions—cont'd

	PREDOMINANT WEAKNESS PATTERN																				
	Proximal > Distal	Dist. Symmetric	Dist. Asymmetric	Scapuloperoneal	Dist. Lower Limb	EOM	Periocular Involve	Bulbar Dysfunction	Contractures	CNS Involvement	Acute Weakness	Hearing Involvement	Exercise Induced	Cardiac Involvement	Resp. Involvement	Myotonia/Cramps	Muscle Pain	Myoglobinuria	CPK > 1000	Immune Mediated	Inflammatory
Congenital Myopathies																					
Central core	X						X									X	X	X	X		
Nemaline	X	X					X								X						
Centronuclear	X	X				X	X								X						
Multicore	X					X	X								X						
Fiber Type Disproportion																					
Reducing body	X					X	X	X							X						
Fingerprint	X						X			X					X						
Cytoplasmic body	X						X								X						
Myofibrillar	X	X					X							X	X						
Metabolic Myopathies																					
Acid maltase deficiency (Pompe disease)	X			X					X	X				X	X		X		X		
Myophosphorylase deficiency (McArdle disease)	X	X	X	X	X								X	X	X	X	X	X	X		
Disorders of lipid metabolism	X										X		X			X		X			
Respiratory chain defects	X												X								
Distal Myopathies																					
Welander					X																
Markesbery-Griggs-Udd					X																
Nonaka			X		X										X						
Miyoshi					X														X		
Laing		X			X									X							
Mitochondrial Myopathies												X						X			
Kearns-Sayre syndrome	X					X								X				X			

Progressive external ophthalmoplegia				X					X									X
Mitochondrial encephalomyopathy, lactic acidosis, stroke (MELAS)	X			X	X			X		X	X	X						X
Myoclonic epilepsy, ragged red fibers (MERRF)	X	X		X	X			X										X
Neuropathy, ataxia, retinitis pigmentosa (NARP)	X	X						X		X								X
Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE)	X			X	X	X		X										X
Leber hereditary optic neuropathy	X									X								X
Leigh syndrome	X							X				X						X
Acquired Myopathies																		
Inflammatory Myopathies																		
Dermatomyositis	X					X	X			X	X		X					X
Inclusion body myositis	X		X			X					X		X					X
Polymyositis	X										X		X					X
Necrotizing myopathy	X										X		X		X	X		
Antisynthetase myopathy	X										X		X		X	X		
Toxic Myopathies																		
Alcohol-related myopathies	X									X				X				
Myopathies related to other medications	X													X				
Corticosteroid myopathies	X													X				
Lipid-lowering agent myopathies	X													X		X	X	
Endocrine Myopathies																		
Muscle disorders w/ glucocorticoid abnormalities	X													X				
Myopathies w/ thyroid disease	X					X		X					X	X		X		

Continued

TABLE
43.1

Inherited and Acquired Myopathies and Their Common Weakness Patterns and Associated Conditions—cont'd

	PREDOMINANT WEAKNESS PATTERN																				
	Proximal > Distal	Dist. Symmetric	Dist. Asymmetric	Scapuloperoneal	Dist. Lower Limb	EOM	Periocular Involve	Bulbar Dysfunction	Contractures	CNS Involvement	Acute Weakness	Hearing Involvement	Exercise Induced	Cardiac Involvement	Resp. Involvement	Myotonia/Cramps	Muscle Pain	Myoglobinuria	CPK > 1000	Immune Mediated	Inflammatory
Myopathies w/ parathyroid disease	X															X	X				
Myopathies related to other medications																	X				
Infectious and Granulomatous Myopathies																					
Viral, bacterial, fungal, tuberculous, and parasitic myositis	X																X				
Sarcoid myopathy	X	X															X			X	X
Myopathies associated w/ systemic disease	X	X															X				
Electrolyte disturbances	X	X											X				X				
CNS, Central nervous system; CPK, creatine phosphokinase; Dist., distal; EOM, extraocular muscle; Involve., involvement; Resp., respiratory.																					

CNS, Central nervous system; CPK, creatine phosphokinase; Dist., distal; EOM, extraocular muscle; Involve., involvement; Resp., respiratory.



• **Fig. 43.1** Scapular winging in patient with facioscapulohumeral muscular dystrophy.



• **Fig. 43.3** Ptosis in patient with oculopharyngeal muscular dystrophy.



• **Fig. 43.2** Calf pseudohypertrophy.



• **Fig. 43.4** Anticipation in mother and daughter with myotonic muscular dystrophy.



• **Fig. 43.5** Percussion myotonia whereby a localized myotonic contraction results from percussion over the abductor pollicis brevis muscle.

Diagnostic Evaluation

Laboratory Testing

Laboratory testing should be guided by the clinical and family history as well as the patient's physical examination. Creatine kinase (CK) is an enzyme found in various body tissues, including muscle, and it is one of the most useful labs to obtain in a myopathy workup. CK levels become elevated in muscle disease as it leaks out from damaged muscle cells into the bloodstream. The CK level therefore often correlates to the degree of muscle damage. CK levels may be elevated in both hereditary and acquired forms of muscle disease secondary to a variety of underlying mechanisms, such as metabolic defects, genetic abnormalities, damage from toxins, and inflammation. CK levels can even become elevated in individuals without muscle disease after strenuous exercise. Elevated CK levels are therefore a nonspecific finding. Aspartate aminotransferase (AST), alanine aminotransferase (ALT), aldolase, and lactate dehydrogenase (LDH) may also be elevated in muscle disease and are often ordered as part of the laboratory evaluation. However, AST, ALT, LDH, and aldolase may all be elevated in hepatic dysfunction as well. A gamma-glutamyl transferase level, which is specific for liver disease, should therefore also be obtained to differentiate underlying pathology. Based on the patient's history, other useful labs may include an electrolyte panel to evaluate potassium and calcium levels, thyroid function tests, inflammatory markers (including erythrocyte sedimentation rate), rheumatologic labs (such as antinuclear antibodies), and serum immunofixation, which may be abnormal in acquired amyloidosis. Specialized panels to look for autoantibodies associated with inflammatory myopathies, such as hydroxymethylglutaryl-coenzyme A (HMG-CoA) reductase antibodies, may also be considered when the history is suggestive. In circumstances where a neuromuscular junction disorder such

as myasthenia gravis is suspected, specific autoantibodies, such as acetylcholine receptor antibodies (among others), should be obtained. Diagnostic tests for select myopathic disorders can be found in [Table 43.2](#).

Electrodiagnostic Testing

Nerve Conduction Studies

Electrodiagnostic testing is key in differentiating neuropathic etiologies from myopathic etiologies in the evaluation of weakness. Sensory nerve conduction studies are usually normal in myopathies, though certain myopathic conditions, such as IBM, can be associated with a polyneuropathy.¹⁶ Motor nerve conduction studies are also usually normal, though compound motor action potential amplitudes can be low in particularly severe or end-stage myopathies affecting distal muscles. Significant delays in onset latencies, reduced conduction velocities, conduction blocks, and temporal dispersion are not characteristic of myopathies and should alert the electromyographer to the possibility of a neuropathy. If a neuromuscular junction disorder is on the differential, techniques such as repetitive nerve stimulation may also be employed to look for evidence of a defect in neuromuscular transmission.

Electromyography

Needle electromyography (EMG) can provide several diagnostic clues through the evaluation of abnormal spontaneous activity in addition to motor unit action potential analysis. Fibrillation potentials, positive sharp waves, and complex repetitive discharges are common in neuropathic conditions; however, these waveforms can also be seen in certain myopathic etiologies, especially inflammatory myopathies. Myotonic discharges are most commonly seen in myotonic dystrophies, though may also be appreciated in toxic, hereditary, and metabolic myopathies. A unique electrodiagnostic feature in metabolic myopathies, such as McCordle disease, is a contracture, which is defined by the perplexing absence of any EMG activity during a muscle contraction.

Motor units in myopathic conditions are classically characterized by low amplitude, short duration, and increased polyphasia. Changes in motor unit morphology are secondary to loss of individual muscle fibers. Early recruitment is another common feature observed in motor unit analysis, defined by an abnormally high volume of motor units for the degree of patient effort. It must be emphasized that the recruitment ratio is normal in early recruitment, and there are simply more motor units than expected for a minimal degree of muscle contraction ([Video 43.2](#)).

It should be noted that certain myopathies, such as IBM, may demonstrate a mixture of motor unit morphology and recruitment. Both long-duration, high-amplitude motor units with reduced recruitment and short-duration, low-amplitude motor units with early recruitment may be observed within the same muscle in some patients with IBM.¹⁶ Of note, steroid myopathy often demonstrates no abnormalities on EMG. Abnormal spontaneous activity is typically not seen in steroid myopathy. Additionally, motor unit analysis may be normal as steroid myopathy predominantly affects type II muscle fibers and EMG assesses mostly type I muscle fibers.

Imaging

Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) can be a useful tool across the spectrum of muscle diseases. It can detect characteristic patterns of muscle involvement, assess disease severity, and help select the

TABLE 43.2 Diagnostic Studies for Suspected Myopathic Disease

LABORATORY STUDIES						
Muscle Enzymes			Endocrine			
Creatine phosphokinase			Hemoglobin A1c			
Aldolase			Thyroid-stimulating hormone			
Lactate dehydrogenase			Parathyroid hormone			
Immunologic			Antibody			
Sedimentation rate			Statin-induced myopathy:			
C-reactive protein			HMG-CoA reductase			
Antinuclear antibody			Inflammatory myopathies:			
			NT5C1A (inclusion body myositis)			
			SS-A/Ro			
			SS-B/La			
			Anti-Jo			
			Rheumatoid factor			
			Anti-double-stranded DNA			
Other						
Complete blood count comprehensive metabolic panel			Metabolic and mitochondrial myopathies:			
Liver function tests			Acyl-carnitine profile			
Gamma-glutamyl transpeptidase			Pyruvate: lactate			
Pompe disease:			Amyloid myopathy:			
Alpha-glucosidase dried blood spot			Monoclonal proteins in serum urine			
Electrodiagnostic Studies						
Nerve conduction study						
Sensory nerve action potential and compound muscle action potential			Typically normal; can see motor axonal loss in severe myopathy			
Electromyography						
Volitional motor unit analysis			Early recruitment			
			Reduced amplitude			
			Short duration; long duration in late stages			
			Polyphasic			
Abnormal spontaneous activity			Normal or irritable, late stage can see fibrillations (fibs) and positive sharp waves			
Neuromuscular junction			Typically normal; can have altered junction in some myopathies such as collagen IV myopathies			
Genetic Testing						
Individual gene test if suspicion is high						
Larger panel if differential remains broad:						
Myopathy/muscular dystrophy next-generation sequencing panel						
Mitochondrial myopathy panel						
Channelopathy panel						
Metabolic myopathy panel						
MUSCLE BIOPSY STAINS						
EM	H&E	PAS	Oil Red/Sudan	NADH	Modified Gömöri	Immunohistochemical Staining
Inclusion body myositis	Fat vacuoles, necrosis	Glycogen and carbohydrate disorders	Mitochondrial myopathy; lipid storage	Muscle fiber internal architecture; cores; myofibrillary and mitochondrial myopathies	Connective tissue; nemaline rods; ragged red fibers	Dystrophin, sarcoglycan, dystroglycan laminin, collagen, emerin, dysferlin
EM, Electromicroscopy; H&E, hematoxylin and eosin; NADH, nicotinamide adenine dinucleotide; PAS, periodic acid–Schiff.						

EM, Electromicroscopy; H&E, hematoxylin and eosin; NADH, nicotinamide adenine dinucleotide; PAS, periodic acid–Schiff.

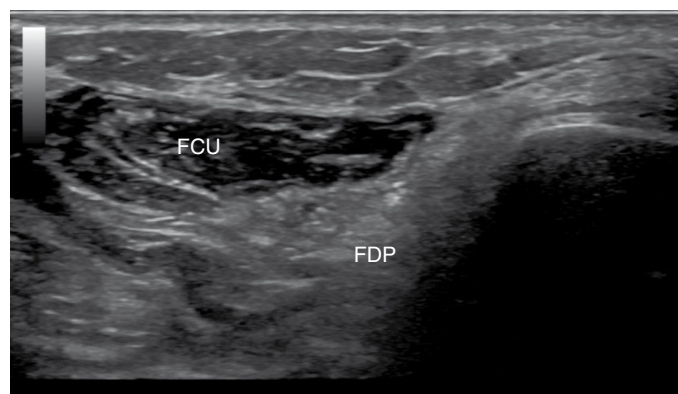
most appropriate biopsy site by identifying areas of active inflammation, necrosis, or abnormal muscle tissue.^{1,29,40} Additionally, MRI is increasingly being used as an outcome measure in clinical trials.⁴⁴ Quantitative MRI techniques, such as muscle volumetry, diffusion tensor imaging, and magnetic resonance spectroscopy, provide objective measures of muscle structure, function, and metabolism, which can be used to assess treatment efficacy and monitor disease progression.

Ultrasound

Some of the earliest research performed using ultrasound in the field of neuromuscular medicine was driven by Heckmatt in the 1980s in the study of muscle disease. Heckmatt et al. described the now famous Heckmatt scale,²⁰ a 4-point scale grading the severity of muscle disease based on the echogenicity. There are clear advantages in using ultrasound over MRI, namely how easily and affordably it can be performed at the bedside. Ultrasound can provide similar benefits to MRI in evaluating muscle disease, including identifying patterns of abnormalities characteristic of specific diseases, assisting in muscle biopsy site selection, and assessing disease severity. Ultrasound can be especially helpful in myopathies with selective muscle involvement, such as IBM, where the flexor digitorum profundus shows increased hyperechogenicity and loss of muscle echotexture when compared to the flexor carpi ulnaris (Fig. 43.6). Ultrasound is less sensitive than MRI in detecting muscle edema, though it can provide similar information about tissue architecture and the presence of pathology.⁵³

Muscle Biopsy

Though less commonly performed in recent decades, muscle biopsy can still provide critical information in certain cases leading to the correct diagnosis. Muscle biopsy can also give a sense of disease severity at times. Muscle biopsy selection site is guided by the clues from the history and examination, surgical accessibility, likelihood of diagnostic yield, and goal of minimizing harm from the procedure. As mentioned earlier in this chapter, imaging with MRI or ultrasound can help identify an appropriate muscle for biopsy. The goal is to obtain a sample from a muscle that clearly demonstrates pathology; however, the muscle selected should not be so advanced in the disease course that it is end stage because severe atrophy and fibrosis can impair analysis. A Medical Research Council strength



• **Fig. 43.6** Muscle ultrasound of the medial forearm in patient with inclusion body myositis. The flexor digitorum profundus (FDP) showed marked atrophy and hyperechogenicity, while the flexor carpi ulnaris (FCU) is relatively spared.

grade of 4/5 is often ideal, whereas a muscle with grade 3/5 is likely too severely affected.¹² Muscles that have been recently studied with needle EMG should be avoided.

The procedure itself is typically performed under local anesthesia. The type of surgeon who performs the procedure will vary by institution, but it is often a general surgeon, neurosurgeon, orthopedic surgeon, or plastic surgeon. In some institutions, nonsurgical physicians, such as neurologists, have privileges to perform the procedure. The procedure can be done either as an open biopsy or needle biopsy. An open biopsy has the advantage of providing a larger specimen for pathologic analysis and is often preferred for that reason.³⁶ The biopsy specimen is sent to pathology for further evaluation. Frozen sections are then processed with various techniques, the choice of which depends on the clinical context and laboratory performing the analysis. A common protocol may involve hematoxylin and eosin staining, histochemistry for Gömöri trichrome, enzyme histochemistry for nicotinamide adenine dinucleotide tetrazolium reductase, succinate dehydrogenase, cytochrome C oxidase/succinate dehydrogenase combination, and fiber typing using slow myosin and fast myosin immunohistochemistry. These studies can help differentiate neurogenic from myogenic pathology and can identify common features of inflammatory myopathies, IBM, congenital/inherited myopathies, vacuolar myopathies, metabolic myopathies, vasculitis, and amyloidosis.³⁶

Forearm Exercise Testing

Forearm exercise testing can be used in the evaluation of metabolic myopathies. A nonischemic forearm exercise test is performed by taking a serum lactate and ammonia level from the antecubital vein before and after effortful hand grip dynamometry protocol.⁴⁸ A significant increase in ammonia levels without an increase in lactate is characteristic of glycogen storage diseases (GSD). Of note, nonischemic forearm exercise testing is preferred over ischemic forearm exercise testing because of the risk of rhabdomyolysis and compartment syndrome with the latter.

Genetic Testing

Advances in genetic testing have provided incredible improvements in diagnostics in recent years. The expense of genetic testing and requirement of prior authorization from insurance can still be a barrier in some cases, though sponsored testing has become more common for certain diseases and incurs no cost to the patient. Patients who have lived with undifferentiated muscle diseases for decades may now have access to testing that can identify a specific etiology. In some cases, genetic testing is all that is needed to confirm a diagnosis, and patients can avoid undergoing uncomfortable procedures, such as electrodiagnostic testing and muscle biopsy. Genetic testing can also provide important information about disease manifestations affecting other organ systems. For example, knowing that certain genetic conditions are also associated with cardiac disease will allow the provider to perform appropriate screening and refer to specialists. Knowing the genetic etiology can also provide important information regarding the patient's prognosis. Patients can also use this information to seek genetic counseling for family planning. Finally, understanding the genetics of specific muscle diseases has profound treatment implications. There are now several US Food and Drug Administration (FDA)–approved gene therapies available for patients with identified mutations (see later).

Specific Diseases

Hereditary Myopathies

Muscular Dystrophies

Muscular dystrophies are a group of hereditary disorders characterized by progressive muscle weakness and dystrophic muscle fibers with fatty replacement on biopsy. Some muscular dystrophies are associated with cardiac or respiratory dysfunction or other organ involvement.³¹ Various mutations causing disruption in muscle structure or function are involved in muscular dystrophies (Fig. 43.7).

Duchenne Muscular Dystrophy and Becker Muscular Dystrophy. DMD and BMD are X-linked recessive diseases caused by mutations in the dystrophin gene. Dystrophin is a cytoskeletal protein, which promotes stability and functionality of muscle fibers. The gene encoding dystrophin is one of the largest coding genes in the human genome, and it is thus prone to mutation. DMD affects approximately 15 cases per 100,000 live male births, whereas BMD is less common. DMD patients have a more severe phenotype. In DMD, the dystrophin gene makes a nonfunctional protein cause by either a frameshift mutation or nonsense mutation creating a premature stop codon.⁴ In BMD, however, there is usually a missense mutation preserving the natural open reading frame, resulting in a truncated protein that is still somewhat functional.

Disease progression in DMD is well characterized. Initial signs begin in early childhood (ages 2 to 6 years) with weakness, clumsiness, difficulty climbing stairs, requiring use of hands to get up from the floor (Gower sign) (Fig. 43.8), and hypertrophied calves.⁴

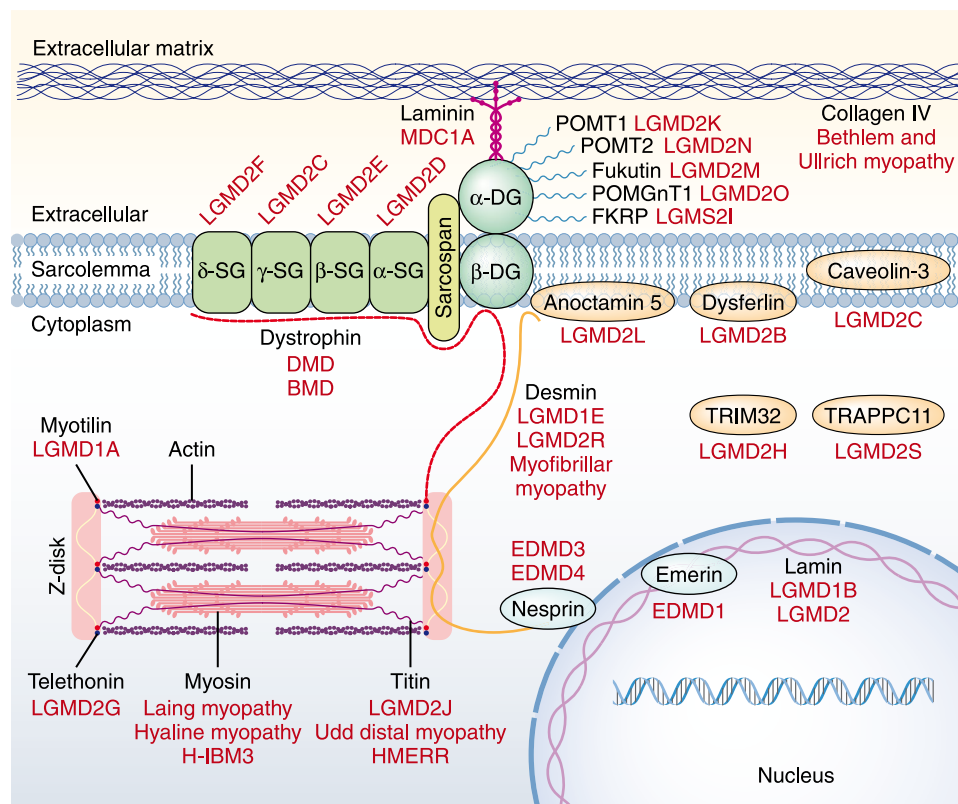
Difficulties with ambulation progress throughout childhood, and the median age for loss of ambulation is 13 to 14 years. After transitioning to a wheelchair for mobility, patients continue to have progressive weakness of the upper extremities and worsening cardiomyopathy and restrictive lung disease. Most patients die from cardiac or pulmonary complications before the fourth decade of life.⁵

Diagnosis. Patients with DMD have significantly elevated CK and aldolase levels. Genetic testing is the primary method of diagnosis, and current methods can detect a mutation in dystrophin in most cases. In the rare cases where genetic testing is not unrevealing, muscle biopsy shows muscle fiber size variation, fibrosis, and abnormal staining of dystrophin.⁴

Family members of a patient with DMD should consider genetic testing. Female carriers have several reproductive choices, including preimplantation genetic diagnosis or prenatal genetic testing. Female carriers are at increased risk of cardiomyopathy and should receive regular screening.

Disease-Modifying Therapies. Although there is no cure for DMD, there are many existing and emerging treatment options focused on slowing disease progression and improving quality of life. DMD is the most researched hereditary muscular dystrophy in terms of active clinical trials and pharmaceutical pipeline. In the past few years, there have been multiple medications approved for DMD that target the upstream genetic dysfunction in dystrophin and the downstream biochemical pathways responsible for muscle degeneration.

Corticosteroids have been used as disease-modifying treatment for DMD for over 30 years and have been shown to slow progression of weakness, delay loss of ambulation, and preserve upper



• **Fig. 43.7** The proteins and genes of the muscle cell involved in select myopathic disorders. (Courtesy Lucas Smith, PhD.)



• Fig. 43.8 Gower sign.

extremity strength. Many DMD patients are started on steroids in early childhood. Glucocorticoids have a significant side effect profile, including hyperglycemia, cataracts, weight gain, osteoporosis, and delayed puberty.⁴⁷ Novel formulations of steroids, such as deflazacort and vamorolone, have shown similar efficacy in terms of preserving muscle strength and prolonging ambulation while demonstrating a better side effect profile compared to traditional steroids.^{17,18} The efficacy of continuing steroids in the late nonambulatory stage is an area requiring further research.

Exon skipping therapies are a novel class of medications that use antisense oligonucleotides to “skip” over mutated exons of the dystrophin gene, theoretically restoring the open reading frame and creating a functional, albeit truncated, protein. These treatments are only available to patients with specific mutations in dystrophin that are amenable to skipping. Eteplirsen (exon 51) and golodirsen (exon 53) are FDA-approved intravenous treatments available to DMD patients with these specific mutations and have shown relative increases in dystrophin levels in clinical trials.^{15,27}

In 2023, the FDA approved Elevidys (delandistrogene moxepar-vec) as the first gene transfer therapy for DMD patients between the ages of 4 and 5 years. Elevidys uses an adeno-associated virus vector to deliver a shortened version of the dystrophin gene to skeletal and cardiac muscles resulting in production of a truncated dystrophin protein (microdystrophin). This is a one-time intravenous medication. Results from the clinical trials show improvement in functional rating scales, time to rise, and 10-m walk test.^{41,55}

Duvyzat (givinostat) is a histone deacetylase inhibitor indicated for individuals with DMD age 6 years and older. Absence of dystrophin results in decreased expression of genes involved in muscle regeneration through histone deacetylase activity.³² In a clinical trial, DMD patients on Duvyzat were shown to have improved four-stair climb assessment compared to placebo. Duvyzat received FDA approval in 2024.

Rehabilitation. DMD has a well-characterized pattern of progression with worsening weakness throughout childhood, loss of ambulation and transition to wheelchair use in teenage years, and ultimately cardiac and pulmonary dysfunction leading to mortality. Rehabilitation management uses our knowledge of natural history to develop specific interventions to improve function and quality of life at each stage. A multidisciplinary care team, including a

physician, physical therapist, occupational therapist, speech therapist, orthotist, and an equipment specialist, is recommended.

Ambulatory Stage. Patients benefit from therapies to improve gait, minimize risk of falls, and assess for appropriate assistive devices. Patients should undergo a daily home stretching program to prevent contractures at the ankles, knees, and hips early on, while adding in stretches for the upper extremities later. Resting ankle-foot orthosis can be used to prevent plantarflexion contractures. Education regarding exercise in muscular dystrophy should be provided, emphasizing aerobic exercise and avoiding eccentric high-resistance exercises and monitoring for overexertion.

Nonambulatory Stage. Transitioning to a wheelchair can be a physically and emotionally challenging process. Most DMD patients benefit from a motorized power wheelchair because of their inability to propel a manual wheelchair. Custom-molded seating can be helpful for axial support and offloading pressure over the sacrum and ischial tuberosities. Power tilt and recline reduce the risk of pressure injuries and help relieve stress over the lower back and hips. A pressure-relieving cushion should be used. Wrist/hand orthosis and knee orthotics, or serial casting, can be used to prevent contractures. Dynamic arm support splints can be helpful for proximal weakness and improve independence in activities of daily living (ADLs).

Cardiac. Individuals with DMD develop cardiomyopathy from dysfunction of dystrophin in cardiac muscle fibers. Involving a cardiologist early in the disease course is recommended. After age 10 years, DMD patients should have cardiac screening yearly with electrocardiogram (ECG) and cardiac imaging. Cardiovascular MRI is superior to echocardiogram in assessing cardiomyopathy, left ventricular dysfunction, and muscle fibrosis. DMD patients are at increased risk for rhythm abnormalities and should undergo regular Holter monitoring. Angiotensin-converting enzyme or angiotensin receptor blockers are first-line therapy with the addition of beta blockers and mineralocorticoid receptor antagonists later in the disease course.⁵ Heart transplantation has been shown to be a viable option for end-stage heart disease in slower progressing muscular dystrophies such as BMD.⁵¹ However, cardiac transplantation in DMD is a complex ethical dilemma because late-stage DMD patients also suffer from significant restrictive lung disease, gastrointestinal dysfunction, and limited functional independence, which often excludes these patients from transplant lists.

Pulmonary. DMD patients develop progressive restrictive lung disease from diaphragm and respiratory muscle weakness and chest wall deformities. Loss of ambulation is associated with decreasing forced vital capacity and onset of pulmonary dysfunction. Early symptoms of respiratory dysfunction include impaired cough reflex, nocturnal hypoventilation, and frequent respiratory infections, while later symptoms include frank dyspnea and orthopnea. Routine pulmonary function testing (PFTs) should be assessed measuring functional vital capacity, maximum inspiratory pressure, and maximum expiratory pressure. Nocturnal hypoventilation can be assessed by asking questions about quality of sleep, fatigue, impaired concentration, nighttime arousals, morning headaches, and the need to take naps throughout the day. Patients with symptoms of nocturnal hypoventilation should be assessed with polysomnography. A weak cough reflex can lead to mucous plugging, atelectasis, or pneumonia. A mechanical insufflation-exsufflation (cough-assist) device gradually inflates the lungs and then rapidly changes to negative pressure to clear secretions. Initiation of noninvasive ventilation (NIV) at night has been shown to improve survival and improve nocturnal hypoventilation. In late-stage disease, as respiratory function decreases, patients will gradually require NIV throughout the day as well.⁵ In the past, patients with muscular dystrophy and restrictive lung disease often received tracheostomies, especially when they were hospitalized in the setting of acute respiratory failure. However, improvements in our understanding of noninvasive pulmonary care, especially the addition of daytime mouthpiece ventilation, has drastically reduced the need for tracheostomies. Most DMD patients with end-stage pulmonary dysfunction can maintain stable vital capacity and normocapnia with NIV.⁵⁰ Tracheostomy can still be considered in some cases, and considerations would include a discussion regarding possible impairments to speech, swallowing, and the need for increased respiratory care.

Gastrointestinal and Nutritional Management. Patients with DMD can have dysphagia, abnormal weight, and nutritional imbalances caused by immobility, decreased energy expenditure, and glucocorticoid therapy. DMD patients should regularly be assessed by a dietician to track intake, nutritional status, and weight. Proper nutritional care is essential to prevent either obesity or malnourishment. Dysphagia typically presents late in DMD. Mild dysphagia can be treated with speech therapy evaluation for swallowing techniques and diet modification. Severe dysphagia requires placement of a gastrostomy tube.

Endocrine Complications. DMD and the use of corticosteroids lead to endocrine complications of impaired growth, delayed puberty, and adrenal insufficiency.

Growth and Puberty. Linear growth should be assessed every 6 months until completion of puberty. For nonstanding patients, height can be approximated segmentally measuring the individual body parts. Patients with a significant decline in growth trajectory or a height less than the third percentile should be referred to endocrinology.

Delayed puberty is a potential complication of glucocorticoid therapy. Patients should be regularly checked for pubertal development, and biochemical testing should be performed for concern for delayed puberty. Testosterone replacement therapy is recommended for confirmed hypogonadism in DMD patients older than 14 years.⁴

Adrenal Insufficiency. Adrenal crisis caused by chronic suppression of the hypothalamic-pituitary-adrenal axis is a potentially life-threatening condition that can develop if glucocorticoids are stopped suddenly. Signs of adrenal crisis include lethargy,

abdominal pain, worsening weakness, decreased appetite, dizziness, fever, dehydration, diarrhea, or hypoglycemia. Stress dosing of steroids may be required for patients with severe illness, major trauma, or surgery. Glucocorticosteroids cannot be stopped abruptly but can be very gradually tapered over a period of months via the PJ Nicholoff Steroid Protocol.²⁵

Bone Health. Treatment with glucocorticoids can lead to osteoporosis, putting DMD patients at higher risk for long bone and vertebral compression fractures. Laboratory testing for calcium, vitamin D, phosphate, magnesium, and parathyroid hormone should be assessed regularly. Dual x-ray absorptiometry scan should be performed to assess bone mineral density; however, compared to healthy individuals, bone mineral density may not be sufficient in evaluating for osteoporosis. Vertebral fractures in DMD patients can often be asymptomatic, and regular thoracolumbar x-rays are recommended even if the patient does not have low back pain. Dietary supplements with calcium and vitamin D are recommended. In patients with severe osteoporosis or a history of long bone or vertebral fractures, treatment with bisphosphonates should be considered.⁴

Orthopedic Complications. In the ambulatory stage, plantarflexion contractures that are negatively impacting ambulation can be considered for Achilles tendon-lengthening surgery. Assessment for scoliosis should be completed at least annually via physical assessment and spinal radiographs. Worsening of scoliosis occurs often after loss of ambulation. Posterior spinal fusion should be considered for spinal curve greater than 20 degrees, especially in patients who are nonambulatory or have not reached puberty.⁴ In the late nonambulatory stage, surgical interventions for contractures are not recommended unless there are issues with pain, positioning, or skin integrity.

Limb Girdle Muscular Dystrophy. LGMDs are a broad group of hereditary disorders involving skeletal muscles, which are characterized by symmetric proximal weakness of shoulder and pelvic girdles. LGMDs have an estimated incidence of 1 to 6 of 100,000, although this figure may be an underestimate given the variability of presentation and underdiagnosis of these disorders.²² Based on the gene involved, clinical symptoms may present at any time from early childhood to late adulthood. Initial weakness is typically in proximal muscles of the arms and legs, although with progression, weakness may spread to more distal muscle groups and/or axial and neck muscles. Some types of LGMD are also associated with cardiac or pulmonary dysfunction. Extraocular muscles and cranial muscles are usually spared.

The increased availability and affordability of molecular genetic testing have drastically changed the diagnostic algorithm in LGMDs. Multigene panels can test large numbers of genes simultaneously. Often genetic testing alone can lead to swift and accurate diagnosis. However, genetic testing can result in variants of unknown significance (VUS), in which there is a mutation in a causative gene; however, it is uncertain whether this mutation sufficiently disrupts protein function. In these cases, further investigations are required. Muscle biopsy shows dystrophic changes with muscle fiber necrosis and regeneration and replacement with fibrosis and fatty tissue. Immunohistologic staining can be helpful. In most LGMD, CK is elevated. EMG of weak muscles should show myopathic motor units.

Some forms of LGMD are inherited in an autosomal dominant pattern, while others are recessive. Recently, there has been considerable debate into the classifications of LGMDs and which diseases qualify as LGMDs. The original classification system for LGMDs was derived in the 1995 by the European Neuromuscular Centre,

which used 1 for autosomal dominant LGMDs and 2 for autosomal recessive LGMDs, followed by a letter based on the order of discovery (e.g., LGMD-2A is a calpainopathy). More recently, in 2018, the same group proposed a new nomenclature of using LGMD with D for dominant or R for recessive followed by a number showing the order in which it was discovered (e.g., LGMD-R1 is calpainopathy).⁴⁶ Furthermore, in the new nomenclature, certain genotypes that were thought to not fit the phenotypical description of LGMD were excluded. For ease of understanding, both nomenclatures are shown in our discussion, and the most common types of LGMDs are described. Next is a list of the most common LGMDs. A comprehensive list of the many mutations resulting in this phenotype is beyond the scope of this chapter.

LGMD-R1/LGMD-2A Calpainopathy. Calpainopathy is the most common form of LGMD worldwide. Age of onset is between 2 and 45 years. The limb girdle pattern of weakness is most common. Scapular winging is common in this disorder, and calpainopathy should be in the differential diagnosis for suspected FSHD. Joint contractures are found in more than one-half of patients, frequently affecting the ankles, hips, knees, and elbows. The heart and lungs are usually not affected.⁵²

LGMD-R2/LGMD-2B Dysferlinopathy. Dysferlinopathy is the second most common subtype of LGMD in the United States; however, in Asia, it is the most common subtype. Age of onset is usually between 15 and 30 years. Dysferlinopathy can present with the typical proximal pattern of weakness, but it also has a predilection for distal lower extremity weakness, particularly in plantarflexion. Calves are usually atrophied. Interestingly, before onset of symptoms, patients with dysferlinopathies tend to be more athletic than their peers, with roughly 20% competing in sports on a regional or national level. Cardiac involvement is rare. Pulmonary complications may arise late in disease course. CK can be markedly elevated ($>30,000$ U/L).⁵²

LGMD-2C(D,E,F)/LGMD-R3(4,5,6) Sarcoglycanopathy. The sarcoglycan proteins (alpha-, beta-, gamma-, and delta-sarcoglycan) form a tetrameric structure in the muscle membrane. Mutation in any of the four sarcoglycan proteins can result in myopathy, and the phenotypes of the four sarcoglycanopathies are similar. Disease onset is usually within the first decade of life, beginning with a typical pattern of proximal upper and lower extremity weakness. Patients often have calf hypertrophy, macroglossia, scapular winging, and scoliosis. Weakness can be severe, and the majority of patients are wheelchair dependent by the time they are teenagers. Dilated cardiomyopathy and restrictive lung disease are common, and patients should have regular screening.

Myofibrillar Myopathy/Previously LGMD-1A. This disease process involves mutations in the gene myotilin and can cause progressive weakness both proximally and distally. Previously classified as LGMD-1A, recent proposals suggest removing this disease from LGMD because of its predilection for distal weakness. The age of onset has a wide variation from teenagers to late 60s and is inherited in an autosomal dominant pattern. Muscle biopsy shows myofibrillar dissolution.⁴³

Emery-Dreifuss Muscular Dystrophy. EDMD is a unique inherited muscular dystrophy characterized by muscle weakness, joint contractures, and cardiac dysfunction. Muscle weakness commonly presents in a humeroperoneal pattern, preferentially affecting the biceps and triceps in the arms and the ankle dorsiflexors and evertors in the legs. In contrast to most myopathies, the shoulders, hips, and thighs are usually spared. Neck weakness and scapular winging are also commonly seen. Contractures frequently present by the first decade of life and classically affect

neck extension, elbow flexion, and ankle plantarflexion. The majority of EDMD patients have cardiac dysfunction, which can range from arrhythmias, syncope, or exercise intolerance to cardiomyopathy and heart failure.²¹

Diagnostic Evaluation. CK levels can range from normal to 15 times the upper limit of normal. Muscle biopsy shows dystrophic features, disruption of myofibrillar architecture, and lack of antiemerin antibodies. Electrodiagnostic studies show EDMD results from defect of one or more proteins in the nuclear envelope, possibly caused by dysfunction of protein importation into the nucleus. Many genes can cause this phenotype, including *EMD*, *LMNA*, *SYNE1*, *SYNE2*, *SUN1*, and *SUN2*.

Management. Physical therapy for flexibility and range of motion (ROM) should be initiated for contractures; however, often these contractures are severe and require surgical intervention. Achilles tendon-lengthening procedures have shown the most efficacy. Elbow flexion contracture surgery can be more complicated and has a lower rate of success. Surgical intervention for neck contractures can be considered, although the risks and benefits must be carefully weighed.

Facioscapulohumeral Muscular Dystrophy. FSHD is the third most common form of muscular dystrophy, resulting from abnormal expression of the *DUX4* gene in skeletal muscle. The two subtypes of FSHD have similar clinical presentations and molecular basis but distinct genotypes. FSHD-1 accounts for 95% of cases, and FSHD-2 makes up the remaining 5%.⁴⁹ It typically presents in the second decade of life, though it may present as early as infancy. It is autosomal dominant, though up to 30% of cases arise from sporadic mutations. FSHD is characterized by facial weakness, scapular winging, and humeral muscle weakness with regional sparing of the deltoids. The pattern of weakness of FSHD may notably be asymmetric. Extramuscular manifestations are less pronounced than in other types of muscular dystrophy and more common in childhood presentations. These manifestations may include respiratory involvement, retinal vascular disease, hearing loss, and possible cardiac abnormalities.

Diagnostic Evaluation. In patients with a classic phenotype and genetically confirmed FSHD in a first-degree relative, a clinical diagnosis is often sufficient. Genetic testing is the definitive diagnostic modality for identifying FSHD. Most protocols start by analyzing the size of D4Z4 repeat array. FSHD-1 results from a D4Z4 repeat contraction and therefore cannot be identified on whole exome sequencing. If no D4Z4 repeat contraction is identified, further testing for FSHD-2 is performed by assessing the degree of methylation of the 4q35 subtelomeric region.³⁴ Serum CK levels may be normal or elevated, but not more than five times the upper limit of normal. Electrodiagnostic testing and muscle biopsy often demonstrate nonspecific myopathic features.

Management. There is currently no validated pharmacologic treatment to treat or slow the progression of FSHD. Engaging in low-intensity aerobic exercise with an experienced physical therapist may be beneficial, although it is important to not overexert the patient to the point of worsening muscle weakness. While respiratory compromise is less common in FSHD, baseline PFTs should be obtained at the time of diagnosis. Those with abnormal PFTs and/or excessive daytime sleepiness (EDS) should have interval monitoring and a referral for a sleep study, as some patients may require nocturnal NIV. Routine cardiac monitoring is not indicated, though any patient with cardiac signs or symptoms should undergo further testing. Those with large D4Z4 repeat contractions may be at risk of retinal vascular disease and

should be referred to ophthalmology. Individuals with childhood-onset FSHD should undergo screening for hearing loss annually.

Oculopharyngeal Muscular Dystrophy. OPMD is a late-onset myopathy characterized by weakness of the eyelid and bulbar muscles. Patients present in the fifth or sixth decade of life with progressive ptosis and dysphagia. In later stages of disease, extraocular movements can be affected leading to diplopia, and there can be mild weakness in the facial, temporal, and masseter muscles. Patients can also have proximal weakness in the shoulders or pelvic girdle. Intrinsic eye muscles and retinal function are preserved. OPMD is caused by GCG expansions of the polyadenylate-binding protein nuclear-1 (*PABPN1*) gene. This disorder usually has an autosomal dominant pattern of inheritance, although there is a rarer autosomal recessive form.

Diagnostic Evaluation. Genetic testing for the expansions in the *PABPN1* gene is the primary diagnostic tool used for suspected OPMD. Laboratory testing can show an elevated CK. Muscle biopsy shows tubulofilamentous intranuclear inclusions in the skeletal muscle. Needle EMG of weak muscles demonstrates a myopathic pattern.⁶

Management. Dysphagia can be evaluated by speech therapy and objectively measured with modified barium swallow test or videofluoroscopic swallowing study (VFSS). Patients should undergo diet modification to the least restrictive diet that can be safely tolerated. Severe weight loss, malnutrition, and aspiration pneumonia are possible complications of untreated dysphagia. Blepharoplasty or frontal suspension of eyelids can be considered for ptosis that is impairing vision.

Distal Myopathies

Distal inherited myopathies present with weakness and atrophy of the distal legs and hands. Symptom onset is variable, but compared to other inherited myopathies, there is more onset likely toward late adulthood. On physical exam, distal myopathies often have preferential weakness of the ankle dorsiflexors compared to the toe extension. This is in contrast to a neuropathic pattern of weakness in which the distal toe extensors are weaker than the dorsiflexors and can be a useful clinical tool in diagnosing distal myopathies.

Welander Distal Myopathy. Welander myopathy is an autosomal dominant late-onset disease with symptoms occurring in the fifth and sixth decades of life. This disease is most often found in the Swedish and Finnish populations. Classically, weakness is found in the long finger extensors and ankle dorsiflexion. There may be mild sensory dysfunction as well. The gene *TIA1* is involved. There is almost no cardiac or pulmonary involvement.¹⁰

Marksberry-Griggs/Udd Late-Onset Distal Myopathy. Marksberry-Griggs distal myopathy typically presents with weakness in the anterior leg compartment in muscles in the fourth to fifth decades of life. Later in the disease course, the distal fingers and proximal arms and legs may be affected as well. Genetic testing shows mutation in the *TTN* gene.¹⁰

Nonaka Distal Myopathy. Nonaka distal myopathy is an autosomal recessive disorder with disease onset in the 20s. Weakness is found preferentially in dorsiflexion and toe extension. Cardiac conduction abnormalities may be present. Mutations in the *GNE* gene are causative.¹⁰

Myotonic Dystrophy

The myotonic muscular dystrophies are multisystem diseases characterized by autosomal dominant repeat expansions. There are two types of myotonic dystrophy. Myotonic dystrophy type 1

(DM1) results from a CTG triplet expansion in the *DMPK* gene, whereas myotonic dystrophy 2 (DM2) results from a CCTG tetramer expansion in the *CNBP* gene.¹⁹ As the name suggests, the myotonic dystrophies are characterized by myotonia. Though patients may describe symptoms that sound similar to muscle cramps, myotonia has distinct pathophysiology and electrodiagnostic features. Myotonia originates at the level of the muscle fiber, whereas muscle cramps are generated at the level of the motor neuron/axon. During needle EMG, myotonic discharges are recognized as spontaneous discharges firing at a rate of 20 to 150 Hz with waxing and waning amplitude and frequency. Myotonic discharges have brief spike or positive sharp wave morphology³⁹ (Video 43.3). Clinically, patients may experience myotonia as delayed muscle relaxation. Patients with grip myotonia, for example, may express difficulty releasing their grasp on a door-knob or after a firm handshake. Myotonia is also exacerbated by cold temperatures.

Myotonic Dystrophy 1. Patients with DM1 have a higher number of CTG repeats (>50) than those without DM1. Early studies of DM1 suggested that a higher number of repeats correlated with greater disease severity and earlier onset, though recent data suggest that other factors, such as downstream sequence interruption and concomitant mutations in other genes, also influence severity. The most severe form of DM1 is congenital, where symptom onset occurs within the first few months of life. Congenital DM1 is often (but not always) characterized by an exceptionally high number of CTG repeats (>750). This occurs most often because of the phenomenon of anticipation, where intergenerational expansion leads to repeat instability and a more severe phenotype. Infants with congenital DM1 may experience hypotonia, respiratory failure, skeletal deformities, and impaired feeding. It is not uncommon for newborns with congenital DM1 to require ventilatory and nutritional support. Those with disease onset between 4 weeks and 10 years of life fall into the category of childhood-onset DM1, and those with onset between 10 and 18 years of life fall into the category of juvenile-onset DM1.¹⁹ The remainder of patients will develop symptoms in adulthood. DM1 preferentially affects muscles of the face, neck/pharynx, trunk, and distal limbs. Facial appearance may appear thin with temporal wasting, and ptosis is common. Patients may experience dysarthria, dysphagia, and difficulty lifting their head off a pillow caused by neck flexor weakness. Proximal muscles may be involved later in the course. Extramuscular manifestations of DM1 include endocrine/metabolic abnormalities, cardiac arrhythmias, cognitive impairment, apathy, EDS, central sleep apnea, impaired gastrointestinal motility, hypersensitivity to anesthetics, increased risk of cancer, and cataracts.

Myotonic Dystrophy 2. DM2 often presents later in life than DM1, with symptom onset typically in the third to fifth decade.³⁹ The average number of CCTG repeats in DM2 is 5000, and there have been no congenital or infantile cases reported. The variability in disease severity in DM2 is not clearly understood. Repeat length in blood cells has not been shown to correlate with disease severity.¹⁹ Patients with DM2 often develop proximal rather than distal muscle weakness. Patients may describe myalgia, stiffness with initiating activity, or difficulty ascending stairs. It is not uncommon for patients with DM2 to go years without the correct diagnosis, which is likely because of milder symptoms, later onset, slower progression, and less specificity in presentation. Despite a milder phenotype than DM1 in general, patients can suffer many of the same extramuscular complications later in the disease course.

Diagnosis. Genetic testing is the diagnostic modality of choice to confirm myotonic dystrophy. In those with typical clinical features and a suspected or genetically confirmed family history, proceeding directly to genetic testing is appropriate. In those with less specific clinical presentation, labs and/or EMG may be performed before genetic testing. CK level may be normal or minimally elevated. EMG is characterized by myotonic discharges and motor units with a myopathic morphology and recruitment pattern in the typical distribution depending on subtype. Muscle pathology often shows nonspecific myopathic features, and it is not required to make the diagnosis.

Management. The consensus care guidelines developed by the Muscular Dystrophy Foundation provide similar guidance in both types of myotonic dystrophy.^{2,42} Patients with myotonic dystrophy can have severe adverse reactions to anesthetics and opioid medications; therefore anesthesia risks should be discussed before surgical or procedural intervention. As pulmonary complications are the leading cause of death in DM1, baseline and annual PFTs should be conducted. Patients with respiratory involvement and/or EDS should be referred to pulmonology. NIV is often recommended for patients with more significant respiratory impairment. Patients with EDS may be counseled about lifestyle interventions and/or prescribed medications to help with alertness, such as modafinil. Cardiac complications are the second leading cause of death in DM1, and ECGs should be obtained annually or biennially. Those with abnormal ECGs or cardiac symptoms should be referred to cardiology. Patients interested in family planning should undergo genetic counseling to understand the risks of inheritance in offspring. Patients planning to become pregnant should also be counseled about the increased risk of complications and be referred to a high-risk obstetrician. Speech, swallow, mobility, fall risk, and independence with ADLs should be evaluated annually with appropriate referrals to respective therapists when indicated. Mexiletine may be considered to manage discomfort or dysfunction secondary to myotonia, though this is contraindicated in patients with cardiac arrhythmias. Patients should undergo annual eye exams to screen for ocular manifestations of myotonic dystrophy, such as cataracts. Patients should be screened for gastrointestinal involvement with appropriate referral to a specialist when indicated. Patients with psychiatric or behavioral manifestations should be referred to an appropriate mental health provider. For metabolic/endocrine manifestations, hemoglobin A1c and liver function tests should be obtained annually. Thyroid function tests and lipid panels should be obtained every 3 years. Sex-specific endocrine abnormalities, such as erectile dysfunction or irregular menstruation, warrant referral to appropriate specialists. Finally, patients with myotonic dystrophy should be counseled about the increased risk of cancer and follow general population cancer screening guidelines.²

Channelopathies: Nondystrophic Myotonias and Periodic Paralysis. Channelopathies are a group of inherited diseases caused by mutations in electrolyte channels (sodium, potassium, and chloride) in the muscle membrane. In the nondystrophic myotonias, such as myotonia congenita (mutations in chloride channel protein *CLCN1*), this can present with muscle stiffness and delayed relaxation after contraction. Clinically, this can be evaluated by having the patient grip tightly for several seconds and assess for delayed relaxation. Sometimes symptoms improve with repeated muscle contractions, in a “warmup” phenomenon. Usually there is no fixed weakness.

Periodic paralysis disorders are characterized by episodes of flaccid muscle paralysis without loss of consciousness. Symptoms

typically start before the age of 20 years. Attacks of episodic paralysis can last from minutes to hours. Mutations in the sodium channel gene *SCN4A* and calcium channel gene *CACNA1S* are associated with periodic paralysis. Triggers can involve temperature, certain carbohydrate-rich foods, stress, and changes in the level of extracellular potassium.⁴⁵ Depolarizing anesthetic agents should be avoided in patients with any channelopathy.

Congenital Myopathies

The congenital myopathies are a group of heterogeneous disorders historically grouped by specific histopathologic features on muscle biopsy. Included in this group are the nemaline myopathies, core myopathies, and centronuclear myopathies. They are characterized by pathology at the level of the contractile matrix, involving structures such as T tubules or the sarcoplasmic reticulum. More recently with advances in genetic testing, several genes of interest have been identified. There is significant variability in time of onset and phenotypic features across the congenital myopathies. Symptoms may include hypotonia, joint contractures and other musculoskeletal deformities, ophthalmoparesis, ptosis, and cardiac involvement, among other manifestations. Patients with features concerning for a congenital myopathy require prompt referral to a pediatric neuromuscular specialist to pursue genetic testing. This often avoids the need for costly or invasive diagnostic studies, such as EMG, muscle biopsy, or advanced imaging. As with many of the myopathic diseases discussed in this chapter, no specific medications exist to treat these diseases, and caring for these patients requires a multidisciplinary approach focused on supportive care and optimization of function.⁷ Table 43.3 shows the common congenital myopathic disorders and their characteristics.

Metabolic Myopathies

Metabolic myopathies are a group of inherited neuromuscular disorders caused by deficiency of enzymes necessary for carbohydrate or lipid metabolism, leading to exercise intolerance, rhabdomyolysis, and sometimes multiorgan involvement. During short bouts of high-intensity exercise, the primary metabolic pathways involve the breakdown of glycogen or anaerobic breakdown of glucose. Thus disorders of carbohydrate metabolism are present within this early phase of exercise. Conversely, with prolonged exercise, the body begins to rely on free fatty acid metabolism. Lipid metabolic disorders present in situations of low-intensity, long-duration exercise.³ Mitochondrial myopathies can present similarly to lipid myopathies.

Diagnosis

Elevation of creatinine kinase and aldolase levels after physical activity are often the first signs of metabolic myopathies. Recent advances and widespread availability of molecular genetic testing have greatly expedited the diagnosis of these disorders. There are commercially available panels that test dozens of genes coding for metabolic enzymes simultaneously from a simple saliva sample. Often the diagnosis can be made from genetic testing alone. However, difficulties arise when there are VUS, in which there is a detected mutation in a causative gene; however, current technology cannot determine whether this mutation would be disruptive to normal protein function. In cases of VUS, muscle biopsies, electrodiagnostic testing, forearm testing, bloodwork for glucose and lipid metabolites, and cycle exercise testing may be indicated.

TABLE 43.3 Characteristics of More Common Congenital Myopathies

	Inheritance Pattern	Genetics	Severity	Key Muscular Findings	Key Extramuscular Findings	Key Histologic Features
Nemaline myopathy	AD	ACTA1	Variable	Ankle dorsiflexion Neck flexion Facial	Preserved IQ EOM spared Respiratory Dysphagia High-arched palate Tented upper lip	Nemaline Rods
Core myopathies	AR	NEB				
	AD	RYR1	Variable	Proximal lower limbs	Malignant hyperthermia	Central cores
Multicore (minicore) myopathy	AD	SELENON/ SEPN1 or RYR1	Variable	Axial Neck flexion	Cardiac Respiratory Oculomotor rare	Cores and minicores
Centronuclear (myotubular) myopathy	X-linked (most common)	MTM1	Severe	Facial	Ocular Cardiac is spared	Myotubules
	AR	BIN1, TTN, RYR	Intermediate	Generalized weakness	EOM spared	
	AD	DNM2	Severe	Neck flexors Facial	EOM	
Congenital fiber type disproportion	AD, AR, X-linked	TPM3, TPM2, RYR1, ACTA1	Variable	Nonspecific	Nonspecific	Type 1 atrophy Type 1 fiber predominance

AD, Autosomal dominant; AR, autosomal recessive; EOM, extraocular muscles.

Disorders of Glycogen Metabolism

Patients with GSD develop exercise intolerance and cramps within minutes of exercise. McArdle disease (GSD-V) is the most common metabolic myopathy and caused by a mutation in the *PYGM* gene encoding for myophosphorylase. McArdle disease and the majority of GSD are inherited via an autosomal recessive pattern. Individuals with McArdle disease experience a “second wind” phenomenon, where after a period of low-intensity exercise there is then a decrease in heart rate and an improvement in exercise tolerance. This is caused by the initiation of fatty acid metabolism. Prolonged exertion results in profound muscle stiffness, temporary muscle contractures, rhabdomyolysis, and renal failure. Many individuals with McArdle disease are diagnosed after repeated bouts of elevated creatinine kinase levels leading to hospitalization. With disease progression, about 50% of McArdle disease patients have fixed weakness of the proximal and axial muscles. Many cases of McArdle disease are diagnosed through genetic testing. The exercise forearm test should show a normal rise in serum ammonia without significant rise in lactic acid. Muscle biopsy shows absent staining for myophosphorylase and accumulation of glycogen in vacuoles.

Disorders of Fatty Acid Oxidation

Lipid myopathies present with exercise intolerance and myalgias with low-intensity, long-duration exercises. Stiffness is less common.

Rhabdomyolysis and elevated CK can also be seen with fasting, illness, cold, or surgical procedures. CPT-II deficiency is the most common lipid myopathy, which can be diagnosed through genetic testing. Laboratory studies show increased long-chain fatty acids on acyl carnitine profiling.

Management

Individuals with metabolic myopathies can engage in exercise safely if progressed in a slow, gradual manner. Dietary modification is key to successful therapy, and individuals with McArdle disease may benefit from a ketogenic diet and taking sugars pre-exercise. For lipid myopathies, the diet should be rich in carbohydrates and low in fats.³

Pompe Disease

Pompe disease is an autosomal recessive disorder characterized by glycogen deposits within the lysosomes of muscle fibers. The causative gene is *GAA*, which encodes alpha-glucosidase, an enzyme catalyzing alpha-1,4 and alpha-1,6 linkages in lysosomal glycogen. Pompe disease can present in either early-onset (infantile) or late-onset (adult) form, and the disease activity is related to the amount of residual enzyme activity in the muscle cells. Diagnosis is made by genetic testing or decreased enzyme activity in dried blood spot testing. Myotonic discharges can be seen in the paraspinal muscles on needle EMG. Early-onset Pompe disease typically presents before the age of 1 year and is characterized by

muscle weakness, enlarged tongue, cardiomyopathy, and failure to thrive. Late-onset Pompe disease is characterized by significant respiratory insufficiency (which can be the presenting symptom) and proximal weakness. Enzyme replacement therapy with alglucosidase alfa should be initiated once the diagnosis is made.²⁶ Often patients require respiratory support.

Mitochondrial Myopathies

Mitochondrial myopathies are a large group of disorders defined by genetic abnormalities resulting in pathologic changes to the structure or function of mitochondria. Mitochondrial dysfunction can negatively impact essentially any organ system, and the nervous and muscular systems are often involved because of their high energy requirements. The specific subtypes of mitochondrial diseases are often eponyms (e.g., Kearns-Sayre syndrome) or acronyms derived from the subtype's common manifestations (e.g., MELAS [mitochondrial encephalopathy, lactic acidosis, and stroke-like syndrome]). In general, the current names that exist to describe mitochondrial diseases do not effectively classify the incredible range of genotypic and phenotypic presentations associated with these disorders. To demonstrate this range in clinical presentation, a few of the major phenotypes are briefly discussed here. Leber hereditary optic neuropathy is characterized by subacute painless vision loss. MELAS is characterized by myopathy, seizure disorders, short stature, cognitive impairments, and stroke-like events in nonvascular territories. Myoclonic epilepsy with ragged red fibers is characterized by epilepsy, short stature, sensorineural hearing loss, cardiac rhythm abnormalities, myopathy, and characteristic ragged red fibers on muscle biopsy. Leigh syndrome and neuropathy, ataxia, and retinitis pigmentosa (or NARP) is characterized by a stepwise decline in neurologic function, peripheral neuropathy, retinitis pigmentosa, ventilatory failure, and characteristic findings on MRI brain showing involvement of central gray matter, cerebellar nuclei, periaqueductal gray, and brainstem. The mtDNA deletion and depletion syndromes, such as progressive external ophthalmoplegia, result in a large range of hepatocerebral, encephalomyopathic, or neurogastrointestinal manifestations. Kearns-Sayre syndrome results in external ophthalmoplegia, retinitis pigmentosa, sensorineural loss, myopathy, and cardiac abnormalities. Pearson syndrome results in death within the first few years of life due to sideroblastic anemia, multiple severe endocrinopathies, and pancreatic failure. Other mitochondrial phenotypes include the POLG spectrum disorders, mitochondrial neurogastrointestinal encephalopathy, BCS1L syndromes, OPA1 disorders, and mitochondrial myopathy caused by thymidine kinase-2 deficiency.⁹ The diagnostic approach requires an in-depth understanding of key phenotypic features; therefore involvement of an experienced neuromuscular specialist is critically important. Evaluation with blood tests, genetic testing, and muscle biopsy may aid in the diagnosis. Unfortunately, there are currently no FDA-approved medications or therapies for mitochondrial diseases, and the evidence to support the use of vitamins and cofactors is lacking.

Acquired Myopathies

Idiopathic Inflammatory Myopathies

The idiopathic inflammatory myopathies (IIMs) are a group of multisystem diseases associated with myositis. These disorders most commonly develop between the fourth and sixth decades of life, though children between the ages of 5 and 15 years may

also develop subtypes of IIMs. The majority of these subtypes disproportionately affect females (ratio 2:1) except for IBM.³³ Historically, polymyositis (PM) was the umbrella term for these disorders. As more was learned about specific clinical features, histopathologic manifestations, and serologic autoantibodies, the term PM was eventually separated into specific diseases, such as antisynthetase syndrome (ASyS) and immune-mediated necrotizing myopathy (IMNM). Muscle enzymes may be normal or dramatically elevated (>20 times the upper limit of normal). Nerve conduction studies are typically normal in IIMs. Needle EMG may show abnormal spontaneous activity (fibrillation potentials, positive sharp waves, complex repetitive discharges, or myotonic discharges) when significant inflammation and/or necrosis is present, decreased insertional activity in advanced disease with significant fibrosis, and myopathic motor units with short duration, polyphasia, low amplitude, and early recruitment. Muscle MRI is typically obtained to distinguish muscle edema from fat infiltration, determine the pattern and extent of muscle involvement, assist in muscle selection for biopsy, and assess the response to therapy. Muscle biopsy remains the diagnostic tool of choice to definitively identify the specific subtype in IIMs. Treatment protocols vary by subtype but typically involve corticosteroids and nonsteroidal immunosuppressant drugs often in combination.³⁰ Both rheumatologists and neuromuscular specialists are often involved in the care of patients with IIMs. Table 43.4 shows the most common inflammatory myopathic disorders and their relevant characteristics.

Dermatomyositis

Dermatomyositis is characterized by proximal muscle weakness and typical dermatologic signs, such as Gottron papules, mechanic's hands, and heliotrope rash. Patients may also experience constitutional symptoms, calcinosis cutis, cardiac involvement, and interstitial lung disease, among other manifestations. Dermatomyositis can be classified into further subtypes based on specific autoantibodies, each showcasing distinct features in terms of muscle weakness distribution and extramuscular symptoms. Common autoantibodies associated with dermatomyositis include anti-Mi-2, anti-MDA5, anti-TIF1-gamma, anti-NXP2, and anti-SAE.²⁸ In some cases, an autoantibody cannot be identified. Dermatomyositis is strongly associated with malignancy, especially among patients with anti-TIF1-gamma and anti-NXP2 antibodies.¹⁴ Muscle MRI often shows patchy muscle edema. Muscle biopsy classically demonstrates perifascicular atrophy, among other features.

Antisynthetase Syndrome

The clinical features of ASyS include myositis, fever, rash, interstitial lung disease, arthritis, mechanic's hands, and Raynaud syndrome. There are eight known antibodies associated with ASyS: anti-Jo1, anti-Ha/YRS, anti-Zo, anti-EJ, anti-PL-7, anti-PL-12, anti-OJ, and anti-KS.²⁸

Immune-Mediated Necrotizing Myopathy

IMNM is most commonly associated with anti-signal recognition particle (anti-SRP) or anti-3-HMG-CoA reductase. Anti-SRP antibodies are associated with a severe and rapid-onset presentation of weakness and CK levels over 10,000. Muscle MRI often shows disproportionate involvement of pelvic and hip adductor muscles. Myocarditis and respiratory failure may develop in anti-SRP IMNM, which often requires aggressive treatment. IMNM associated with anti-HMG-CoA reductase antibodies may present

TABLE 43.4 Idiopathic Inflammatory Myopathies

	Age of Onset	Sex	Muscle Pathology	Creatine Kinase Level		Weakness Pattern	Dermatologic Findings	Responsive Immunotherapy
Dermatomyositis	Childhood and adulthood	Females > males	Perimysial and perivascular inflammation CD4+ T cells	Normal	Elevated	Proximal > distal	Present	Yes
Antisynthetase syndrome	Adult	Females > males	Perifascicular necrosis	X	Variable	Proximal > distal	Absent	Yes
Polymyositis	>20	Females > males	Endomysial inflammation CD8+ T cells MHC-1 antigen expression	X	Up to 50×	Proximal > distal	Absent	Yes
Necrotizing autoimmune myositis	Adult and elderly	Females > males	Necrotic fibers with macrophages CD8+ T cells		Up to 10×	Proximal > distal	Absent	Yes
Inclusion body myositis	>50	Males > females	Rimmed vacuoles; endomysial inflammation CD8+ T cells TDP-43	X	Up to 15–20×	Proximal > distal Asymmetric Forearm/finger flexors Knee extensors Dysphagia	Absent	No

acutely, subacutely, or even chronically. CK tends to be elevated but to a lesser degree than what is seen in anti-SRP-associated IMNM. A history of statin use is observed in about 66% of patients with anti-HMG-CoA reductase IMNM, and cancer is commonly associated with this subtype.³⁰

Inclusion Body Myositis

IBM is the most common acquired muscle disease affecting individuals in the second half of life. The classic phenotype involves atrophy of the deep forearm flexors and quadriceps. As the disease advances, often all skeletal muscles become involved. Aspiration pneumonia secondary to dysphagia is a leading cause of mortality in IBM. It is often misdiagnosed as a different IIM, which can delay the diagnosis by several years. Some patients may have positive anti-cytosolic 5'-nucleotidase 1A antibodies, although studies have shown that this antibody is also present in other neuromuscular diseases. EMG often shows a combination of neuropathic and myopathic features. Key features on histopathology include the invasion of nonnecrotic myofibers by lymphocytes, rimmed vacuoles, and major histocompatibility complex I and II overexpression. Unlike the aforementioned IIMs, IBM does not improve with immunosuppressive therapies.¹⁶

Endocrine Myopathies

Low vitamin D levels (<30 nmol/L) can lead to elevated serum CK levels, exercise intolerance, and rhabdomyolysis. Both hypo- and hyperthyroidism can predispose patients to exertional rhabdomyolysis. Hypothyroidism has also been associated with proximal

myopathies. In fact, up to 80% of patients with hypothyroidism report some degree of muscle cramping, muscle fatigue, or weakness.¹³ Myopathy secondary to corticosteroids can occur because of both exogenous (prescription medications) and endogenous (adrenal tumors, ectopic adrenocorticotrophic hormone production, and pituitary adenomas) sources. Patients typically have other features of Cushing syndrome in addition to muscle weakness, including hirsutism, impaired glucose metabolism, central obesity, purple striae, and neuropsychiatric features, among other signs. Excessive growth hormone (acromegaly) is also known to lead to a progressive proximal myopathy. Though significantly less common, it should be noted that diabetes, gonadal dysfunction, and carcinoid syndrome can rarely lead to myonecrosis, muscle fatigue, or myopathy.²⁴

Toxic Myopathies

The clinical manifestation of toxic myopathy varies widely. Although some patients experience severe symptoms shortly after starting the medication responsible, others may develop mild symptoms gradually over several months of exposure. Cholesterol-lowering medications, cyclosporine, propofol, labetalol, and alcohol can all lead to necrotizing myopathies. Presentation may include proximal muscle weakness, hyperCK-emia, an irritable myopathy on EMG, and occasionally renal involvement. Patients often recover once the offending agent has been identified and the patient is no longer exposed. It is important to note that statins can lead to either a toxic necrotizing myopathy or IMNM (see earlier). Patients diagnosed with the latter will not recover with medication discontinuation alone,

while patients without the immune-mediated component often improve within several weeks to a few months after stopping the medication. There is a higher risk of statin-induced necrotizing myopathy in patients with older age, renal failure, or concomitant use of medications that increase the muscle toxicity of statins (such as cyclosporine).

Immune checkpoint inhibitors, D-penicillamine, cimetidine, phenytoin, interferon alfa, tumor necrosis factor inhibitors, imatinib, and hydroxyurea are associated with inflammatory myopathies. Amiodarone, chloroquine, and hydroxychloroquine are associated with amphiphilic myopathies, which occur because of the disruption of lysosomal function. Colchicine and vincristine may lead to antimicrotubular myopathies through the disruption of microtubules.

Recreational drugs can lead to various muscle pathologies. Repeated intramuscular injection of drugs can lead to focal muscle fibrosis. Individuals who use cocaine, phencyclidine, or amphetamines may experience rhabdomyolysis. Rhabdomyolysis may also occur with compartment syndrome secondary to immobility related to drug use.¹¹

Infectious Myopathies

Infection by bacteria, viruses, fungi, or parasites is a less common etiology of muscle disease, though it is critically important to recognize. The presence of fever, malaise, leukocytosis, and elevated inflammatory markers should prompt consideration of an infectious source. A clinical history of spending time in tropical regions, recent infection elsewhere in the body, penetrating injuries, recent surgery, intramuscular drug use, immunosuppression, muscle ischemia, or exposure to unsanitary conditions may all raise suspicion for infectious myositis. A myriad of pathogens are capable of infecting muscle. Viruses and parasites often lead to diffuse myalgia or multifocal myositis, while bacteria tend to cause focal involvement. Diagnosis relies on laboratory investigations, cultures and/or muscle biopsies, and imaging. Consultation with an infectious disease specialist may be indicated. Treatment with appropriate antimicrobials can prevent severe complications, such as sepsis.³⁵

Critical Illness Myopathy

CIM is a myopathy that can develop in severely ill, hospitalized, or intensive care unit (ICU) patients. CIM can occur with or without critical illness polyneuropathy. CIM can be a significant factor in delayed recovery and prolonged mechanical ventilation. Risk factors for CIM include prolonged ICU stay, mechanical ventilation, severity of acute disease course, sepsis, multiorgan failure, parenteral nutrition, and administration of corticosteroids and aminoglycosides.^{54,56} The underlying mechanisms for CIM are poorly understood but may involve abnormal depolarization of muscle membrane from systemic insult.

The diagnosis of CIM includes (1) a positive history for critical illness with multiorgan dysfunction, (2) clinical limb weakness or difficulty in weaning patient from ventilation, and (3) exclusion of other medical and neurologic causes for increased weakness. Electrodiagnostic testing shows small-amplitude, short-duration motor units with early recruitment with or without fibrillation potentials and positive sharp waves. Muscle biopsy shows myopathy with necrosis. Aggressive treatment of sepsis and decreasing doses of corticosteroids and neuromuscular blocking agents may decrease likelihood of CIM. Early mobilization with physical therapy is recommended when appropriate in ICU patients.

General Rehabilitation Principles in Myopathic Disorders

Despite the incredible range of clinical severity and pathology of the disorders discussed in this chapter, there are several common rehabilitation principles that can be applied when implementing a care plan for patients with muscle disease. As is the case with other chronic conditions managed by physiatrists, the approach to care often involves an interdisciplinary model. It is common for care teams to include physical therapists, occupational therapists, speech therapists, rehabilitation psychologists, nutritionists, social workers, assistive technology representatives, orthotists, and wheelchair vendors. Representatives from volunteer organizations, such as the Muscular Dystrophy Association, may also be present in interdisciplinary clinics. Physician team members may include physiatrists, neurologists, rheumatologists, orthopedic surgeons, endocrinologists, pulmonologists, and cardiologists. Principles of care are further discussed by topic.

Mobility and Exercise

A strength and mobility assessment should take place at each visit, ideally in conjunction with a physical therapist. The distribution and severity of weakness are highly variable across muscle diseases; therefore some patients may require a single-point cane for assistance in ambulation, while others may need a power wheelchair for all mobility. Transfer boards, transfer benches, and lifts facilitate safe and comfortable transfers between different surfaces (e.g., bed to wheelchair, wheelchair to toilet) for individuals with muscle weakness or limited mobility. Physical therapy can help recommend the appropriate mobility aids and collaborate with prosthetics when bracing (such as an ankle-foot orthosis) is recommended.

Overall, there is sparse literature about how to best provide guidance on exercise in patients with muscle disease. However, some care guidelines around exercise do exist for specific disorders. For example, the American Academy of Neurology's evidence-based guidelines for care in FSHD report that low-intensity exercise appears to be safe and beneficial for these patients.⁴⁹ When specific guidelines do not exist, an exercise program may still be considered with input from other members of the care team. A physical therapist may be able to help develop an individualized exercise program for the patient while considering important factors such as the patient's fall risk. Eccentric exercises can exacerbate muscle damage and are not recommended in diseases such as DMD. In general, exercise that results in delayed-onset muscle soreness is not recommended by most neuromuscular physicians. When active movement at a joint is significantly decreased or no longer possible, passive ROM should be employed regularly to avoid contracture development.^{4,5}

Contractures

Contractures can lead to further impairments in function, skin breakdown, poor posture or positioning when seated, pain or discomfort, and difficulty reaching certain body regions for hygiene. ROM throughout the extremities should be regularly assessed both in the doctor's office and by the patient's caregiver(s) at home. Because contractures are quite difficult to treat, it is best to do what is possible to prevent their development to begin with. Frequent passive ROM exercises, stretching, and bracing can all help prevent contractures from developing. Patients may benefit

from custom nocturnal foot splints and/or resting hand splints. If contractures have already developed, management options depend on the affected body region, the patient's age, and the patient's individual situation. Surgical interventions for addressing contractures are often better suited to the pediatric population, though certain adults may also benefit from a referral to orthopedic surgery to discuss options.

Activities of Daily Living and Equipment

Outside of mobility aids, patients with muscle disease often require prescriptions for several other kinds of equipment. Understanding which equipment is needed comes from inquiring about the level of assistance the patient requires to complete ADLs. Collaborating with occupational therapy and/or an assistive technology specialist to identify tools to maximize a patient's independence is helpful. Upper extremity splints, adaptive utensils, universal cuffs, voice-activated technology, and forearm orthoses are just some examples of tools that can be used to optimize a patient's function.

Specialized seating systems, cushions, and positioning aids help individuals with muscle disease maintain proper posture, alignment, and comfort while sitting or lying down. Grab bars, shower chairs, raised toilet seats, and other bathroom safety aids improve accessibility and safety in the bathroom for individuals with muscle weakness or balance issues. Speech-generating devices, communication boards, and adaptive keyboards can assist individuals with muscle weakness affecting speech or fine motor skills in communicating effectively. Home adaptations such as ramps, stairlifts, widened doorways, and grab bars may be necessary to improve accessibility and safety for individuals with muscle disease who have mobility limitations.

Pain Management

Pain is highly prevalent in individuals with muscle diseases. One study of DMD patients stated that 73% of study participants reported pain. There are often multiple contributors to pain in muscle disease, and a comprehensive history and physical exam are required to differentiate possible etiologies. Patients with diseases such as myotonic dystrophy may describe painful cramps, which can be treated with medication (e.g., mexiletine). More often, however, pain in muscle diseases often stems from musculoskeletal etiologies. Progressive weakness can lead to altered biomechanics and posture, which may eventually put significant strain on joints, ligaments, tendons, or muscles. Impaired mobility can also lead to joint contractures, which may cause discomfort and stiffness. After a thorough history and examination, imaging with ultrasound, radiographs, or MRI may be useful to further assess the patient's regional complaint. Bone pain should raise concern for pathologic fractures in patients with osteopenia or osteoporosis. Treatment may include bracing, therapies, non-steroidal antiinflammatories, seating adjustments, or procedural intervention when indicated.³⁷

Dysphagia and Secretion Management

Oropharyngeal dysfunction can occur in various types of muscle diseases. Patients may experience coughing or choking on food, frank aspiration events, or sialorrhea. Patients at risk for swallowing disorders should be asked about these symptoms at each follow-up visit. Those with a positive history should be referred to a

speech therapist to undergo VFSS. During the test, the patient drinks liquids mixed with barium contrast to evaluate the safety and efficiency of swallowing and the bolus movement. Speech therapy often relies on these study results to make recommendations about oral intake, such as thickening liquids. In some cases, study results will show that the patient is at a high enough risk of aspiration that it is not recommended to take any nutrition by mouth. In these cases, a risk-benefit discussion with the patient about taking food and liquids by mouth vs. enteral nutrition should take place. Patients with dysphagia should also be screened for respiratory dysfunction because these symptoms are often present together. Severely impaired respiratory function may preclude patients from undergoing procedures for enteral tube placement; therefore these discussions should occur early in the disease course if possible. Sialorrhea can often be managed with anticholinergic medications such as glycopyrrolate, though patients may begin to experience the opposite problem of dry mouth. Injecting botulinum toxin into the salivary glands can significantly reduce saliva production, and this may be considered on a case-by-case basis depending on the patient's specific disease and comorbidities (see Chapter 3).

Nutrition

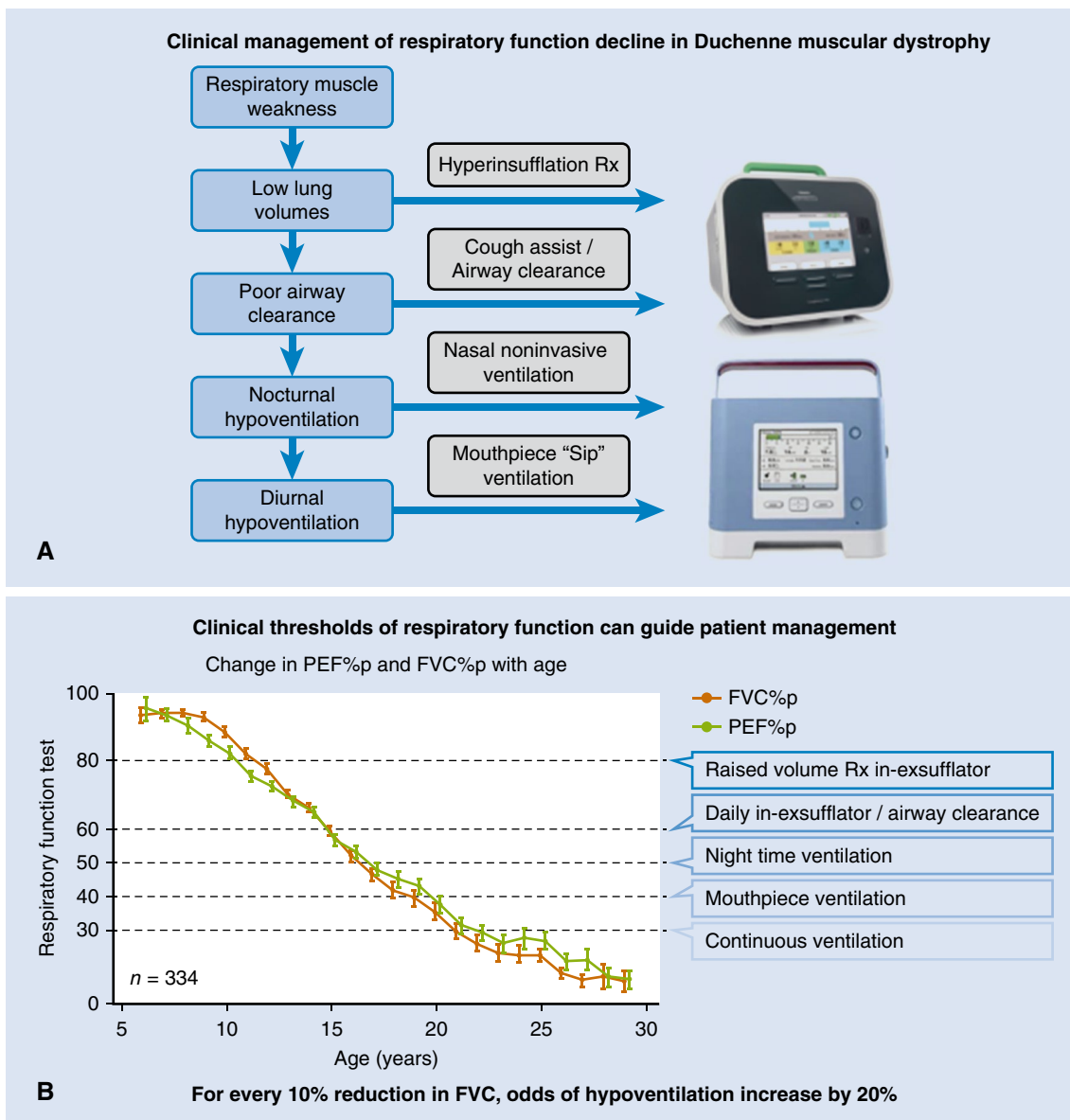
Dietitians are important team members in providing comprehensive care for patients with muscle disease. During their initial assessment, they will obtain information regarding the patient's current nutritional status, dietary intake, and body composition. Depending on the specific disorder, some patients may need assistance with gaining weight while others (such as DMD patients on steroids) may need assistance in avoiding weight gain. Guidance from a dietitian is especially important for those with enteral feeding to ensure the patient is receiving enough essential nutrients.

Cognitive Impairment and Neuropsychiatric Manifestations

Several of the disorders discussed in this chapter are associated with cognitive impairment or psychiatric comorbidities. For example, DM1 is well known to be associated with impaired cognition and apathetic or anxious personality traits.⁸ Attention-deficit/hyperactivity disorder, autism spectrum disorders, anxiety, depression, and obsessive-compulsive disorder are more common in patients with DMD and BMD than in the general population.³⁸ It is important to address mood and cognitive impairment in the patient's history periodically to identify affected patients. Talking with family members or caregivers (if permitted by the patient) can also be helpful in identifying behaviors or impairments that the patient may be unaware of. Depending on the patient's presentation, referrals to psychiatry, speech therapy, neuropsychiatric testing, and/or psychology may be useful.

Respiratory Care

Neuromuscular weakness leading to restrictive lung disease is one of the most serious manifestations in myopathic disorders. It is important to work closely with a respiratory therapist and obtain routine PFTs in those at risk for respiratory involvement. A referral to pulmonology is also warranted early in the course. Patients often require NIV, such as bilevel positive airway pressure, later in the disease course (Fig. 43.9).



• **Fig. 43.9** (A) Clinical management of respiratory function decline in myopathies. (B) Clinical thresholds of respiratory function can guide patient management. *FVC*, Forced vital capacity; *FVC%p*, percent predicted forced vital capacity; *PEF%p*, percent predicted peak expiratory flow. (Data from Birnkrant DJ, Bushby K, Bann CM, et al. Diagnosis and management of Duchenne muscular dystrophy, part 2: respiratory, cardiac, bone health, and orthopaedic management. *Lancet Neurol.* 2018;17(4):347-361; Mayer OH, Henricson EK, McDonald CM, et al. Advances in pulmonary care in Duchenne muscular dystrophy. *US Neurol.* 2017;13(1):35-41.)

Summary

It is an exciting time for providers who treat myopathic disorders. Advances in genetic testing have enhanced diagnostic accuracy and improved our understanding of the specific pathogenic mechanisms involved in myopathic diseases. This has led to an increase in clinical trials and the development of novel therapeutics to slow disease progression and improve quality of life. Multidisciplinary rehabilitation care remains paramount, especially as patients who have traditionally pediatric diseases grow and transition into adult care. The physiatrist is uniquely equipped to provide comprehensive care for this patient population in this new and exciting landscape.

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